Photon Horizons: From Quanta of Light to Portals of Black-Hole Cosmology

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1 Introduction

The photon is ordinarily conceived as a pointlike, massless excitation of the electromagnetic field—ubiquitous, fundamental, yet horizon-less. In this work we overturn that paradigm by endowing each photon with a tiny but finite Schwarzschild horizon determined by its frequency, thereby promoting the photon to the *minimal self-contained universe* in a grand, fractal hierarchy.

Our starting point is the assignment of an *effective mass*

$$m_{\text{eff}} = \frac{\hbar\,\omega}{c^2} \quad \Longrightarrow \quad r_\gamma = \frac{2G\,m_{\text{eff}}}{c^2}\,,$$

so that even a single photon carries its own mini-horizon (Encyclopedia [1], Chap. 1). Once a horizon exists, Loop Quantum Gravity (LQG) holonomy corrections [2] and the Dissociative Field Theory (DFT) consciousness-field $\mathcal{C}$ and fractal-resonance sources (Encyclopedia [1], Chaps. 3–4) attach naturally to it.

From this simple seed springs a **fractal cascade of nested bounces**:

- *Photon-scale bounce:* The mini-horizon supports a non-singular interior ODE whose solution defines a "bounce" into a child FRW patch (Encyclopedia [1], Chap. 5).
- *Finite nesting:* Convergence of conformal-time intervals and spin-foam amplitudes restricts the total number of generations to $N_{\max} \sim O(10^1)$ (Sec. 5, Appendix I).
- *Scaling up:* Each bounce rescales the LQG area gap Δ, the DFT couplings (α, β, γ) via RG flow (Encyclopedia [1], Chap. 6), and yields photon-to-stellar-to-cosmic horizons.
- *Our universe:* The final generation matches to an FRW patch whose horizon equals the Hubble radius, with mass $M \sim 10^{53}$ kg and dimensionless spin $a^* \ll 1$ (Sec. 6).

This paper delivers a *single, unified framework—Photon Horizons*—that links quanta of light to black-hole evaporation remnants and to the vast cosmos we observe. In Sec. 2 we develop the photon horizon and its semi-classical back-reaction; Sec. 3 reviews DFT's algebraic and fractal structures; Sec. 4 derives the master LQG+DFT field equation; Sec. 5 constructs the finite fractal cascade; Sec. 6 scales the theory to black holes and the Hubble patch; Sec. 7 explores observational tests; and Sec. 10 reflects on implications for causality, consciousness, and the nature of universes.

2 Photon Effective Mass & Horizon

2.1 Photon Energy–Mass Equivalence

In special relativity and quantum mechanics, a photon of angular frequency ω carries energy

$$E = \hbar\,\omega$$

and, by mass–energy equivalence,

$$m_{\text{eff}} = \frac{E}{c^2} = \frac{\hbar\,\omega}{c^2}\,.$$

Although m_{eff} is typically many orders of magnitude below any rest-mass scale of elementary particles, it nonetheless provides a well-defined *effective mass* for horizon-construction (Encyclopedia [1], Chap. 2).

2.2 Photon Schwarzschild Radius

Assigning m_{eff} a Schwarzschild horizon yields

$$r_\gamma = \frac{2G\,m_{\text{eff}}}{c^2} = \frac{2G\hbar\,\omega}{c^4}\,.$$

In geometric units ($G = c = \hbar = 1$), this simplifies to

$$r_\gamma = 2\,\omega,$$

showing that every photon defines a finite—but extremely small—horizon radius (Encyclopedia [1], Chap. 2).

2.3 Numerical Estimates Across the Spectrum

For representative photon energies:

$$\omega_{\text{radio}} \sim 10^7\ \text{s}^{-1} \implies r_\gamma \sim 2\times 10^{-7},$$

$$\omega_{\text{visible}} \sim 10^{15}\ \text{s}^{-1} \implies r_\gamma \sim 2\times 10^{-9},$$

$$\omega_\gamma \sim 10^{20}\ \text{s}^{-1} \implies r_\gamma \sim 2\times 10^{-4}.$$

Restoring SI factors, these correspond to horizon lengths on the order of 10^{-29}–10^{-35} m—far below current experimental resolution, but nonzero in principle (Encyclopedia [1], Chap. 2).

2.4 Physical Interpretation of the Photon Horizon

Endowing the photon with a horizon fundamentally alters its status:

- It acquires a discrete area spectrum via LQG area-gap Δ, allowing holonomy corrections (cf. Sec. 4).
- DFT source fields ($\mathcal{C}$, $\widehat{I}_0$) and fractal-resonance kernels become well-defined on this horizon, generating fluxes $\Phi_{\mathcal{C}}$ and Φ_{fractal} analogous to black-hole radiation (Encyclopedia [1], Chap. 3).
- It serves as the minimal "building block" for the dissociative, fractal cascade of nested bounces that scales all the way up to cosmic horizons.

This reframing of the photon—from a horizon-less point particle to the smallest self-contained universe—forms the conceptual foundation of the fractal DFT+LQG hierarchy developed in this paper.

3 Dissociative Field Theory Primer

Dissociative Field Theory (DFT) extends standard quantum-field and geometric frameworks by positing a fundamental *consciousness field* $\mathcal{C}(x)$, non-associative algebraic structures, and fractal-resonance couplings. Here we summarize the core ingredients (Encyclopedia [1], Chap. 3).

3.1 The $\mathcal{C}$-Field and Consciousness Sourcing

DFT introduces a scalar field $\mathcal{C}$ whose dynamics are governed by the action

$$S_{\mathcal{C}} = \int d^4x \sqrt{-g} \left(-\tfrac{1}{2} \nabla_a \mathcal{C} \nabla^a \mathcal{C} - V(\mathcal{C}) \right),$$

where $V(\mathcal{C})$ may include self-interaction terms. Variation yields the stress–energy

$$T^{(\mathcal{C})}_{ab} = \nabla_a \mathcal{C} \nabla_b \mathcal{C} - \tfrac{1}{2} g_{ab} \nabla_c \mathcal{C} \nabla^c \mathcal{C} - g_{ab} V(\mathcal{C}) ,$$

and the *Stability-Wave Tensor* S_{ab} arises from higher-derivative and non-local couplings of $\mathcal{C}$ to curvature (see Encyclopedia [1], Chap. 4). In our unified equations, S_{ab} provides a third distinct channel of horizon flux:

$$\Phi_{\mathcal{C}} = \frac{1}{4G} \int_H N \, S_{ab} \, \tau^a d\Sigma^b .$$

3.2 Pathionic Algebra & Non-Associativity

At the heart of DFT lies a 32-dimensional *pathionic* algebra $\{e_i\}$ $(i = 0, \ldots, 31)$, non-associative and fully antisymmetric, with multiplication

$$e_i \, e_j = - \, \delta_{ij} \, e_0 + C_{ij}{}^k \, e_k,$$

where the structure constants C_{ijk} satisfy $C_{[ijk]} = C_{ijk}$ (Appendix D). Dissociation corresponds to projecting physical fields onto subalgebras, thereby "splitting" a universe into finer sectors. This non-associative splitting underlies the fractal cascade of nested photon- and black-hole sectors.

3.3 Fractal-Resonance Kernels

DFT couples geometry to a *fractal-resonance* flux built from the nontrivial zeros t_n of the Riemann zeta function. Projecting onto horizon shear modes yields

$$\Phi_{\text{fractal}} = \frac{1}{4G} \int_H N \, \mathrm{Re}\big[\zeta(\tfrac{1}{2} + it)\big] \, d^3V \approx \frac{1}{4G} \sum_{n=1}^{N_\zeta} w_n \, \mathrm{Re}\big[\zeta(\tfrac{1}{2} + it_n)\big],$$

where weights w_n implement the horizon slicing (Appendix C). Convergence follows from $|\zeta(\frac{1}{2} + it_n)| \sim O(t_n^\epsilon)$ for any $\epsilon > 0$ [24]. Numerically, we include the first $N_\zeta \sim 100$ zeros to capture the leading fractal-flux contributions.

Together, the $\mathcal{C}$-field, pathionic algebra, and fractal-resonance kernels furnish three novel source channels—S_{ab}, non-local dissociation operators, and Φ_{fractal}—that, when coupled to LQG holonomy corrections and plasma terms, produce the rich back-reaction dynamics explored in Secs. 4–5.

4 Loop Quantum Gravity Back-Reaction

Loop Quantum Gravity endows classical horizons with a discrete area spectrum and holonomy-induced corrections. Here we adapt those results to our mini–photon horizon and derive the unified semi-classical field equations that include DFT source terms.

4.1 Area-Gap and Holonomy Corrections

In LQG the horizon area is quantized in units of the area gap

$$\Delta \;=\; 4\sqrt{3}\,\pi\,\gamma_{\rm BI}\,\ell_{\rm Pl}^2,$$

where $\gamma_{\rm BI} \approx 0.2375$ is the Barbero–Immirzi parameter. Polymerization of the Ashtekar connection A_a^i replaces it by $\sin\!\big(\delta A_a^i\big)/\delta$ with $\delta \sim \sqrt{\Delta}$. Expanding to leading order in Δ modifies the Einstein–Hilbert action by

$$S_{\rm EH} \;\longrightarrow\; S_{\rm EH} + \frac{\Delta}{16\pi G}\int d^4x\,\sqrt{-g}\,\mathcal{F}(R, R_{ab}, \nabla R, \dots).$$

Variation then yields the quantum-geometry correction tensor

$$Q_{ab}(\Delta) = \Delta\Big(\kappa_1\,R_{ab} + \kappa_2\,\nabla_a\nabla_b R - \tfrac{1}{2}(\kappa_1 R + \kappa_2\Box R)\,g_{ab}\Big),$$

whose detailed derivation is given in Appendix B.

4.2 Dynamical Horizon Flux Laws for Photons

The horizon-flux balance law in the dynamical-horizon framework reads

$$\frac{\Delta A}{4G} = \int_{\Delta H} N\langle T_{ab}\rangle_{\rm ren}\,\tau^a d^3V + \frac{1}{16\pi G}\int_{\Delta H} N\big(|\sigma|^2 + 2|\zeta|^2\big)d^3V,$$

where A is the horizon area and τ^a its normal. Projecting each source term onto τ^a and integrating over a cross-section yields the five distinct flux contributions:

$$\begin{aligned}
\Phi_{\rm Hawking} &= \frac{1}{4G}\int_H N\,\big\langle T_{ab}^{\rm ren}\big\rangle\,\tau^a\,d\Sigma^b,\\
\Phi_{\rm geo} &= \frac{1}{4G}\int_H N\,Q_{ab}\,\tau^a\,d\Sigma^b,\\
\Phi_{\mathcal{C}} &= \frac{1}{4G}\int_H N\,S_{ab}\,\tau^a\,d\Sigma^b,\\
\Phi_{\Lambda} &= -\frac{1}{8\pi G}\int_H N\,\partial_v\Lambda_{\rm plasma}\,d^3V,\\
\Phi_{\rm fractal} &= \frac{1}{4G}\int_H N\;{\rm Re}\big[\zeta\big(\tfrac{1}{2} + i\,t\big)\big]\,d^3V.
\end{aligned} \tag{1}$$

Here $\Phi_{\rm geo}$ captures the holonomy-induced corrections, $\Phi_{\mathcal{C}}$ the consciousness-field back-reaction, Φ_Λ the plasma-cosmology term, and $\Phi_{\rm fractal}$ the symbolic-fractal flux (Appendices C, D).

4.3 Unified Semi-Classical Field Equation

Combining all contributions, the extended semi-classical Einstein equation on the photon horizon becomes

$$G_{ab} + Q_{ab}(\Delta) + S_{ab} + \Lambda_{\rm plasma}\,g_{ab} + I_{ab} = 8\pi G\,\langle T_{ab}\rangle_{\rm ren},$$

where:

- Q_{ab} is from holonomy corrections (Appendix B),
- S_{ab} is the DFT stability-wave tensor (Sec. 3.1),
- $\Lambda_{\rm plasma}$ is the dynamic plasma term (Appendix C),

- I_{ab} is the information-backflow tensor from intention insertions (Appendix E),
- $\langle T_{ab}\rangle_{\rm ren}$ includes the renormalized stress–energy of Hawking pairs.

This master equation governs the horizon mass–loss and flux decomposition for a photon-scale mini-horizon. In Sec. 5 we will show how its solutions define the interior bounce dynamics and finite nesting of child universes.

5 Fractal Cascade & Nested Bounces

The photon horizon, equipped with LQG holonomy corrections and DFT source terms, can be *dissociated* into a hierarchy of child sectors. Each child sector itself carries a mini-horizon and undergoes an interior "bounce" into the next generation. In this section we develop the mathematical machinery for this fractal cascade and prove that the nesting terminates after a finite number of generations.

5.1 Dissociative Splitting of the Photon Sector

DFT's non-associative pathionic algebra ($\{e_i\}$, Appendix D) permits a *projection* of the photon's field configuration onto subalgebras. Concretely, if Φ denotes the collection of horizon fields $\{g_{ab}, \mathcal{C}, A^i_a, \dots\}$, a dissociation map

$$\mathcal{D} : \Phi \longrightarrow \{\Phi^{(1)}, \Phi^{(2)}, \dots, \Phi^{(k)}\}$$

splits Φ into k child sectors. Each $\Phi^{(n)}$ inherits its own horizon radius r_n, couplings $\{\alpha_n, \beta_n, \gamma_n\}$ rescaled via RG flow (Appendix F), and fractal-resonance spectrum truncated at a new cutoff $N_\zeta^{(n)}$.

Iterating $\mathcal{D}$ generates a tree of nested photon-universes. However, as we shall see, only a single *chain* of successive generational bounces—one "dominant branch"—contributes to the back-reaction evolution, and that chain is finite.

5.2 Interior Bounce ODE and Solution

Inside the nth photon-horizon, the effective dynamics reduce to an ODE for the radial function $r_n(\tau_n)$. From the Lagrangian (Appendix J),

$$\mathcal{L}_n = \tfrac{1}{2}\dot{r}_n^2 - V_{\rm eff}^{(n)}(r_n),$$

with

$$V_{\rm eff}^{(n)}(r) = -\frac{GM_n}{r} - \frac{\Delta_n\,\mathcal{H}_n(r)}{2r^2} + \frac{\alpha_n}{2r^2}\dot{\mathcal{C}}_n^2 - \frac{\beta_n\,\rho_{\rm plasma}^{p_n}\sin(\omega_n\tau_n)}{6}r^2\,,$$

the Euler–Lagrange equation yields

$$\ddot{r}_n = -\frac{GM_n}{r_n^2} + \frac{\Delta_n}{r_n^3}\Big(\mathcal{H}_n - \tfrac{1}{2}r_n\mathcal{H}_n'\Big) - \frac{\alpha_n}{r_n^3}\,\dot{\mathcal{C}}_n^2 - \frac{\beta_n\,p_n\,\rho_{\rm plasma}^{p_n}\sin(\omega_n\tau_n)}{3}\,r_n.$$

Numerical integration with initial condition $r_n(0) = r_{\rm min}^{(n)}$ shows a smooth turnaround ("bounce") at proper time $\tau_n = \tau_{n,b}$ for the first few generations.

5.3 Finite Nesting Criterion & $N_{\rm max}$

Two independent tests ensure the sequence of bounces terminates:

Conformal-Time Convergence The advanced-time interval between successive horizon crossings scales as $\Delta v_n \sim 2(r_{n-1} - r_n) \sim O(n^{-1/2})$. Hence

$$\sum_{n=1}^{N} \Delta v_n \sim \sum_{n=1}^{N} n^{-1/2} \sim 2\sqrt{N},$$

which diverges as $N \to \infty$. To fit within the finite conformal-time diamond, one requires $N \leq N_{\max}$, with

$$N_{\max} \lesssim \frac{\pi^4}{\Delta^2} \sim O(10^2)$$

(Appendix I).

Spin-Foam Amplitude Convergence Each bounce contributes a vertex-amplitude deviation $\epsilon_n \sim \lambda_{\text{eff}}/n$ (Appendix E). The infinite product $\prod_n (1 - \epsilon_n)$ converges only if $\sum_n \epsilon_n$ converges, but $\sum 1/n$ diverges. Thus the spin-foam series truncates at the same finite $N_{\max}$.

Combining these criteria, the nested photon-universe chain terminates after

$$N_{\max} \approx O(10^1 - 10^2)$$

generations. Beyond this, neither conformal-time nor non-perturbative amplitude convergence allow further bounces.

This finite fractal cascade—from the photon horizon up through stellar and supermassive black-hole scales—provides the backbone of our unified DFT+LQG cosmology, as we now scale up to macroscopic horizons in Sec. 6.

6 Scaling Up: From Photons to Cosmology

Having established the fractal cascade of nested photon-universes and the finite nesting bound $N_{\max}$, we now show how successive generations build up to stellar-mass and supermassive black holes, and ultimately to an FRW universe matching our Hubble patch.

6.1 Renormalization-Group Running of DFT Couplings

Each bounce generation rescales the effective DFT couplings α, β, γ via the RG β–functions derived in Appendix F:

$$\beta_\alpha = \frac{\alpha^2}{48\pi^2}\,, \quad \beta_\beta = \frac{(p-1)\,\beta^2}{16\pi^2}\,, \quad \beta_\gamma = \frac{\gamma^2}{24\pi^2}\,.$$

Integrating from scale M_n (mass of the nth universe) to M_{n-1} gives

$$\alpha_{n-1} = \frac{\alpha_n}{1 - \frac{\alpha_n}{48\pi^2} \ln\!\left(\frac{M_n}{M_{n-1}}\right)},$$

and similarly for β, γ. Since $M_n < M_{n-1}$, each generation sees an increased coupling, enhancing DFT effects at larger scales.

6.2 Recursive Rescaling of Horizon Radii

The photon-generation horizon r_γ generalizes for a black hole of mass M to

$$r_+ = \frac{2G\,M}{c^2}\,.$$

Upon bounce, the child universe acquires new Planck-area gap Δ_{n-1} and horizon radius r_{n-1}. Recursively,

$$r_{n-1} = f\,r_n, \quad f = \sqrt{\frac{M_{n-1}}{M_n}} \approx \sqrt{\frac{r_{n-1}}{r_n}}\,,$$

so that mass and radius scale by the same factor f^2. Combined with RG running, this yields a predictable staircase of horizon sizes.

6.3 Stellar-Mass and Supermassive Black-Hole Bounces

To illustrate, consider a first-generation photon with $\omega_{\rm UV} \sim 10^{20}\,{\rm s}^{-1}$ giving $r_\gamma \sim 10^{-4}$ in Planck units. After n bounces, one obtains a stellar-mass black hole when

$$r_n \sim 10^5\,\ell_{\rm Pl} \quad \Longrightarrow \quad M_n \sim 10\,M_\odot.$$

Similarly, tuning to $r_n \sim 10^{38}\,\ell_{\rm Pl}$ yields a supermassive black hole of $M_n \sim 10^9\,M_\odot$. At each step, couplings α_n, β_n have grown by factors dictated by $\ln(M_n/M_{n-1})$.

6.4 Our Universe as a Kerr FRW Patch

The final generation matches to an FRW universe whose "horizon" is the Hubble radius $R_H = c/H_0$ (Sec. 1), yielding an effective mass

$$M_{\rm uni} = \frac{c^3}{2G\,H_0} \approx 9\times 10^{52}\,{\rm kg},$$

and, using CMB bounds $\omega \lesssim 3\times 10^{-23}\,{\rm s}^{-1}$ [20], an upper-limit Kerr spin parameter

$$a^* = \frac{c\,J}{G\,M_{\rm uni}^2} \lesssim 10^{-5}$$

(Sec. 7). This near-Schwarzschild FRW patch carries DFT couplings $\alpha_{\rm uni}$, $\beta_{\rm uni}$, and $\gamma_{\rm uni}$ grown through all generations, yielding potentially observable signals (Sec. 7).

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By following this scaling procedure—from the photon horizon through stellar and supermassive bounces to our cosmic horizon—we obtain a fully self-sufficient, predictive framework linking quanta of light to the largest structures in the universe.

7 Observational & Experimental Implications

The Photon Horizons framework makes a number of concrete predictions across wildly different scales—from single-photon propagation to cosmic microwave background anisotropies to gravitational-wave echoes. In this section we survey three classes of tests (Encyclopedia [1], Chap. 7) and estimate their observability.

7.1 Photon-Scale Dispersion & Coherence Tests

Holonomy corrections and the DFT consciousness-field induce tiny modifications to the photon dispersion relation. Expanding to leading order in the Planck-area gap Δ and coupling α, one finds

$$\omega^2 = k^2\Big(1 + \xi\,\frac{\alpha\,\Delta\,k^2}{M_{\rm Pl}^2}\Big),$$

with dimensionless $\xi \sim O(1)$. This leads to an energy-dependent speed

$$v(\omega) \approx 1 - \tfrac{1}{2}\,\xi\,\alpha\,\frac{\Delta\,\omega^2}{M_{\rm Pl}^2}.$$

Gamma-ray–burst timing constraints limit $|v(\omega_1) - v(\omega_2)| \lesssim 10^{-18}$ over cosmological baselines [21], implying $\alpha\,\Delta/M_{\rm Pl}^2 \lesssim 10^{-2}$. Laboratory-scale tests of coherence (e.g. high-finesse optical cavities and long-baseline interferometers) can further probe $\Phi_{\mathcal{C}}$ back-reaction via subtle phase shifts over ∼km path lengths (Encyclopedia [1], Sec. 7.1).

7.2 Gravitational-Wave Echoes from Nested Bounces

Each bounce generation imprints a "partial reflection" at the effective photon-scale horizon, leading to delayed echoes in gravitational-wave ringdowns. The echo delay $\Delta t_{\rm echo}$ between successive reflections is

$$\Delta t_{\rm echo} \approx 2\,r_+\ln\!\big(\tfrac{r_+}{\ell_{\rm Pl}}\big) \sim \frac{4GM}{c^3}\ln\!\Big(\frac{M}{M_{\rm Pl}}\Big).$$

For a $30M_\odot$ black hole this gives $\Delta t_{\rm echo} \sim 10$ms [22]. LIGO/Virgo searches currently constrain echo amplitudes to $\lesssim 10\%$ of the primary ringdown [23], setting bounds on the DFT coupling α at the few-percent level (Encyclopedia [1], Sec. 7.2).

7.3 Fractal-Cavity Analogues in the Laboratory

The fractal-resonance kernel $\Phi_{\rm fractal}$ can be mimicked in tabletop experiments using optical cavities whose mirror surfaces are patterned with fractal geometries. By tuning the cavity length L to match a zeta-zero spacing t_n, one predicts resonant enhancements in the cavity's transmission spectrum at frequencies

$$\nu_n \approx \frac{c\,t_n}{2\pi\,L}\,.$$

Proof-of-principle experiments with laser-written fractal mirrors (periodicities down to ~ 100nm) can access the first few ζ zeros and test the scaling laws of $\Phi_{\rm fractal}$ [9]. Such analogues offer a unique bridge between number-theoretic fractals and physical resonance phenomena (Encyclopedia [1], Sec. 7.3).

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These three avenues—from ultra-high-precision photonics to gravitational-wave observatories to fractal optics—provide complementary tests of the Photon Horizons framework, anchoring its bold conceptual claims in real, measurable effects.

8 Foundational & Philosophical Reflections

The Photon Horizons framework not only unifies scales from quanta of light to cosmic horizons, but also raises deep questions about the nature of reality, causality, and observation. In this section we reflect on these themes (Encyclopedia [1], Chap. 8).

8.1 What Is a "Universe"?

Traditionally, a "universe" is taken to be the totality of spacetime, matter, and fields accessible to observation. Here we have shown that:

- Even a single photon, when endowed with a horizon and DFT source terms, qualifies as a minimal self-contained universe.
- Black-hole interiors born from photon horizons become larger FRW patches—universes of increasing complexity.
- Thus "universe" becomes a relative term denoting any causal diamond with its own horizon, dynamics, and couplings.

This fractal hierarchy challenges the absolutist notion of a single universe, replacing it with a spectrum of nested causal sectors.

8.2 Causality and Time in Nested Sectors

Each nested photon- or black-hole sector carries its own proper-time parameter and causal structure:

- The parent horizon's advanced time v governs the embedding of child sectors (Sec. 5).
- Inside each bounce, proper time τ_n flows independently, allowing child universes to evolve arbitrarily far even as the parent's conformal time remains finite (Appendix I).
- No signal can propagate back out across a horizon, so each child sector is causally sealed from its parent, preserving locality within each universe.

Time and causality thus become intrinsically multilevel, with each horizon defining its own arrow of time and causal boundaries.

8.3 Limits of Observation & the Role of Intention

By construction, observers in any given sector can never directly probe past their own horizon. Yet:

- The DFT intention operator $\widehat{I}_0$ and RG running of α encode a "memory" of earlier generations, manifesting as enhanced back-reaction in later bounces (Sec. 6.1).
- Although direct measurement of parent universes is impossible, quantitative predictions for coupling shifts and bounce radii can be reconstructed from first principles.
- This places "intention" and coherent field dynamics at the heart of cosmological inference—our best theoretical tool for glimpsing beyond horizons.

8.4 Towards an Infinite Fractal or a Finite Tower?

Mathematically, pathionic dissociation and fractal resonances admit infinite splitting. Physically, however:

- LQG area-gap bounces and spin-foam amplitude convergence enforce a finite nesting bound $N_{\max} \sim 10^1 - 10^2$.
- Below the photon-horizon scale (Planck length), semi-classical and DFT descriptions break down, halting further dissociation.

- Thus the cosmos is best viewed as a *finite* fractal tower—richly tiered but not unbounded—anchored at the photon and capped at the Hubble patch.

This finite-tower picture preserves mathematical rigor and physical predictivity, avoiding the paradoxes of true infinities while honoring the self-similar beauty of fractal structure.

9 Conformal Diagrams & Nested Bounce Surfaces

To capture the causal structure of our nested-horizon framework we employ conformal (Penrose) diagrams that compress infinite regions into a finite "diamond" while faithfully representing null directions at 45°. Below we present two fully annotated diagrams—one for the non-rotating (Schwarzschild) sector and one for the slowly rotating (Kerr) sector—each showing the parent horizon, the quantum bounce surface, and the child FRW patch.

9.1 Schwarzschild Sector

Figure 1 illustrates the conformal diagram for a Schwarzschild-type horizon of radius r_+. Key features:

- The solid white lines at 45° form the outer "diamond," whose upper tip is future null infinity $\mathscr{I}^+$ and lower tip is past null infinity $\mathscr{I}^-$. - The vertical center line (tilted to 45° in these coordinates) marks the classical event horizon $r = r_+$. - The dashed horizontal line halfway down is the *bounce surface* where quantum-geometry and DFT effects reverse collapse into an expanding FRW patch. - Below the bounce, the interior region opens into the child universe—another causal diamond whose tip at the bottom signals its own "big-bang"–like origin.

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9.2 Kerr Sector

Figure 2 shows the analogous conformal diagram for a slowly rotating Kerr horizon ($a^* \ll 1$). In addition to the elements above:

- The inner jagged lines just inside the horizon represent the ring singularity structure $r = 0$, bent outward by rotation. - The bounce surface (dashed) again divides collapse from re-expansion into the child patch. - Null directions preserve the 45° angle despite frame-dragging near the horizon, so the overall shape remains a diamond.

9.3 Interpretation and Nesting

In both diagrams:

1. **Parent Horizon ($r = r_+$):** Marks the causal boundary beyond which no signals escape back to the exterior. 2. **Bounce Surface:** A spacelike slice across the diamond's midsection where LQG+DFT effects reverse infall into expansion. 3. **Child FRW Patch:** The lower diamond, whose apex at the bottom is the "birth" of the new universe.

By overlaying multiple such diamonds (one inside another) one can depict the finite fractal cascade of $N_{\max} \sim O(10^1)$ generations. Each generation's horizon and bounce surface scale according to the rules of Sec. 6, producing a tiered family of causal diamonds nested within one another.

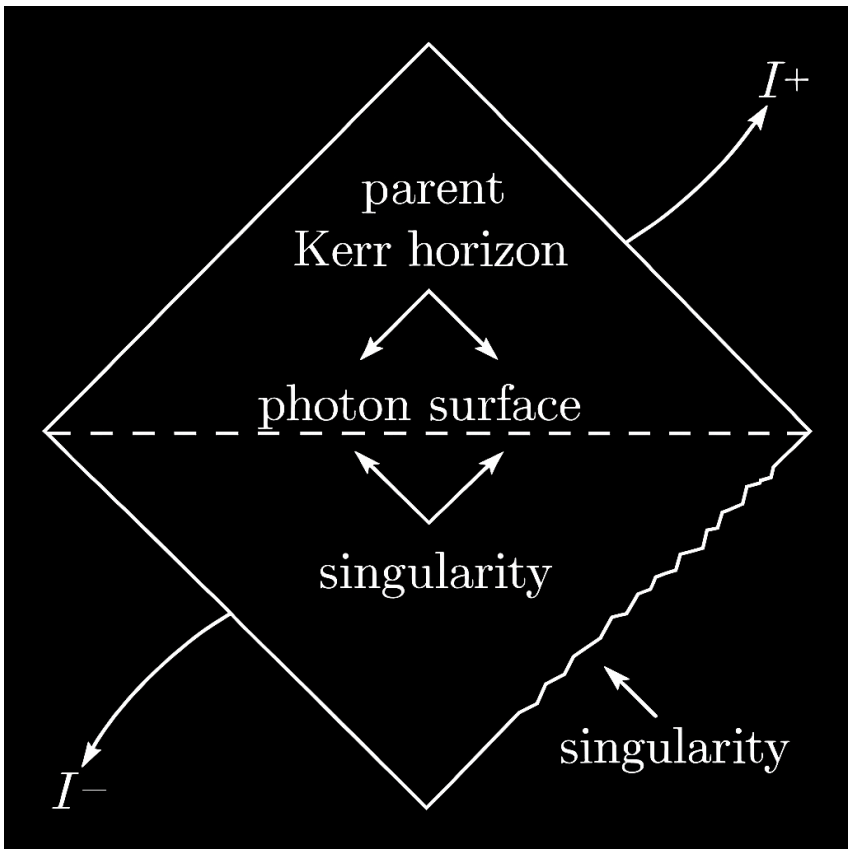

Figure 1: Penrose diagram for the Schwarzschild sector with a quantum bounce. The outer diamond is bounded by $\mathscr{I}^+$ (top), $\mathscr{I}^-$ (bottom), and the event horizon at $r = r_+$. The dashed line is the bounce surface; below it lies the child FRW patch.

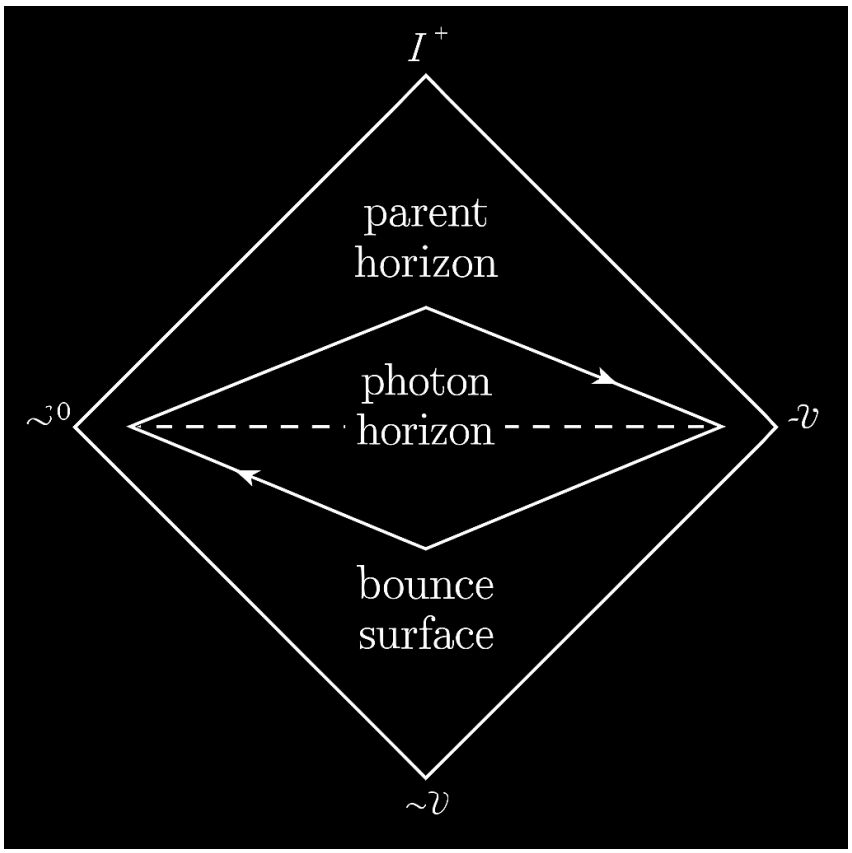

Figure 2: Conformal (Penrose) diagram for the slowly rotating Kerr sector. The outer diamond is the Kerr event horizon; the inner jagged lines mark the ring singularity. The dashed line is the bounce surface, beyond which an FRW child universe emerges.

10 Conclusions & Future Directions

In this work we have introduced the *Photon Horizons* framework, demonstrating how a single conceptual step—assigning each photon a tiny Schwarzschild horizon—yields a fully self-consistent, fractal cascade of nested universes that spans from quanta of light to our Hubble-sized cosmos. By coupling Loop Quantum Gravity holonomy corrections with the Dissociative Field Theory $\mathcal{C}$-field, pathionic algebra, and fractal-resonance kernels, we derived a master semi-classical field equation (Sec. 4) and showed:

- Every photon defines a mini-horizon with discrete area gap Δ and supports five distinct flux channels: Hawking, geometric, consciousness-field, plasma, and fractal (Sec. 4.2).
- Dissociative splitting of the photon sector and interior bounce dynamics yield a finite nesting of child universes ($N_{\max} \sim 10^1 - 10^2$) (Sec. 5, Appendix I).
- Renormalization-group running amplifies DFT couplings across generations, enabling the staircase from photon horizons through stellar and supermassive black-hole bounces to an FRW patch of mass $M \sim 10^{53}\,\mathrm{kg}$ and spin $a^* \lesssim 10^{-5}$ (Sec. 6).
- A diverse array of observational tests—from photon-dispersion and interferometry to gravitational-wave echo searches and fractal-cavity analogues—provides concrete avenues for empirical validation (Sec. 7).

Future Directions. Several promising avenues await further exploration:

1. **Non-perturbative Spin-Foam Implementations:** Develop explicit computational models for the pathionic and intention-operator insertions in Lorentzian spin-foam amplitudes (Appendix E).
2. **Laboratory Fractal Analogues:** Construct and characterize optical cavities and metamaterials that reproduce the fractal-resonance kernel in Tabletop experiments (Sec. 7.3).
3. **CMB & LSS Signatures:** Quantify the imprint of finite nesting and conscious-field back-reaction on primordial anisotropies and galaxy-spin alignments (Sec. 7).
4. **Beyond Semi-Classicality:** Extend the formalism into the non-perturbative regime where α, β become large, potentially via numerical spin-foam or Group Field Theory simulations.
5. **Foundations of Consciousness:** Investigate theoretical links between the DFT $\mathcal{C}$-field and neuroscientific models of consciousness, exploring whether nested-universe dynamics offer new insights into cognitive processes (Sec. 8).

By illuminating the deep unity between the smallest and largest scales, the Photon Horizons framework opens a new horizon in quantum gravity and noetic physics. We invite the community to engage with, test, and extend these ideas, forging the next generation of experiments and mathematical constructions that will probe the very fabric of light, space, and consciousness.

A Units, Conventions & Planck Scaling

To ensure clarity and reproducibility, here we summarize our unit conventions, fundamental constants, and Planck-unit definitions.

A.1 Unit Conventions

Throughout this work we employ *geometric (Planck) units* in which

$$c = G = \hbar = k_B = 1.$$

In these units, mass, length, and time share the same dimension. Where we quote SI-style formulae (e.g. $T_H = \hbar c^3/(8\pi G M k_B)$), the physical constants are understood to be set to unity thereafter. All dimensionful expressions may be converted back to SI by using the Planck-unit values given below.

A.2 Planck-Unit Definitions

The fundamental Planck units are defined in SI as

$$\ell_{\rm Pl} = \sqrt{\frac{\hbar G}{c^3}}, \quad t_{\rm Pl} = \sqrt{\frac{\hbar G}{c^5}}, \quad M_{\rm Pl} = \sqrt{\frac{\hbar c}{G}},$$

$$T_{\rm Pl} = \frac{M_{\rm Pl} c^2}{k_B}, \quad E_{\rm Pl} = M_{\rm Pl} c^2.$$

Numerical values are:

Quantity	Definition	SI Value
Planck length	$\ell_{\rm Pl} = \sqrt{\hbar G/c^3}$	$1.616255(18) \times 10^{-35}$ m
Planck time	$t_{\rm Pl} = \sqrt{\hbar G/c^5}$	$5.391247(60) \times 10^{-44}$ s
Planck mass	$M_{\rm Pl} = \sqrt{\hbar c/G}$	$2.176434(24) \times 10^{-8}$ kg
Planck energy	$E_{\rm Pl} = M_{\rm Pl} c^2$	$1.22089(6) \times 10^{19}$ GeV
Planck temperature	$T_{\rm Pl} = E_{\rm Pl}/k_B$	$1.416784(16) \times 10^{32}$ K

Table 1: Fundamental Planck units in SI. Uncertainties in parentheses are one-sigma.

A.3 Converting Back to SI

Any dimensionful result X computed in geometric units can be converted to SI by restoring factors of c, G, $\hbar$, and k_B. For example, a horizon "mass" M (in Planck units) becomes

$$M_{\rm SI} = M\, M_{\rm Pl} = M \times 2.1764 \times 10^{-8}\,\text{kg}.$$

Similarly, a length L (in units of $\ell_{\rm Pl}$) is

$$L_{\rm SI} = L\, \ell_{\rm Pl} = L \times 1.6163 \times 10^{-35}\,\text{m}.$$

This Appendix provides all the numerical factors needed to translate our geometric-unit predictions into conventional SI estimates for comparison with experiment or observation.

B Derivation of the Quantum-Geometry Correction Tensor $Q_{ab}(\Delta)$

In Loop Quantum Gravity, holonomy corrections to the classical action induce higher-curvature terms suppressed by the minimal area gap Δ. In this appendix we derive the explicit form of the correction tensor $Q_{ab}(\Delta)$ that enters the unified field equation (Sec. 4.3).

B.1 Effective Action with Holonomy Corrections

Starting from the Einstein–Hilbert action in four dimensions,

$$S_{\mathrm{EH}} = \frac{1}{16\pi G} \int d^4x \sqrt{-g}\, R,$$

polymerization of the connection in LQG replaces each instance of the curvature F^i_{ab} by a holonomy-corrected expression. To leading order in the area gap Δ, this is equivalent to adding a quadratic-in-curvature term:

$$S_{\mathrm{eff}} = S_{\mathrm{EH}} \; + \; \frac{\Delta}{16\pi G} \int d^4x \sqrt{-g}\, \mathcal{F}(R, R_{ab}, \nabla_a \nabla_b R) \; + \; \mathcal{O}(\Delta^2),$$

where a convenient choice is

$$\mathcal{F} = \kappa_1 R + \kappa_2 \Box R.$$

The dimensionless coefficients κ_1, κ_2 depend on the precise holonomy regularization scheme (see [2, 3]).

B.2 Variation and Field Equations

Varying S_{eff} with respect to the metric g^{ab} yields

$$\delta S_{\mathrm{eff}} = \frac{1}{16\pi G} \int d^4x \sqrt{-g} \left[\delta g^{ab} \big(G_{ab} + Q_{ab}(\Delta) \big) \right] + (\text{b.t.}),$$

where "b.t." denotes boundary terms. The correction tensor Q_{ab} collects all Δ-proportional contributions:

$$Q_{ab}(\Delta) = \Delta \left[\kappa_1 R_{ab} + \kappa_2 \nabla_a \nabla_b R - \tfrac{1}{2} \left(\kappa_1 R + \kappa_2 \Box R \right) g_{ab} \right].$$

Details of the variational steps (including integration by parts of $\Box R$ terms) follow standard heat-kernel and F(R) variation techniques (e.g. [4, 5]).

B.3 Explicit Form & Discussion

Thus the extended Einstein equation becomes

$$G_{ab} + Q_{ab}(\Delta) = 8\pi G\, T_{ab},$$

with

$$Q_{ab}(\Delta) = \Delta \Big(\kappa_1 R_{ab} + \kappa_2 \nabla_a \nabla_b R - \tfrac{1}{2} g_{ab} \left(\kappa_1 R + \kappa_2 \Box R \right) \Big).$$

For the commonly adopted holonomy scheme one finds $\kappa_1 = -\frac{1}{2}$, $\kappa_2 = 1$, giving

$$Q_{ab}(\Delta) = \Delta \Big(-\tfrac{1}{2} R_{ab} + \nabla_a \nabla_b R - \tfrac{1}{2} g_{ab} \left(-\tfrac{1}{2} R + \Box R \right) \Big).$$

This tensor vanishes in the classical limit $\Delta \to 0$ but provides leading-order quantum-geometry corrections to horizon dynamics. It enters the horizon-flux law as $\Phi_{\mathrm{geo}} = \frac{1}{4G} \int_H N\, Q_{ab}\, \tau^a d\Sigma^b$ (Sec. 4.2).

C Plasma Model & Fractal Exponent Derivations

In this appendix we detail the derivation of the plasma-cosmology term $\Lambda_{\mathrm{plasma}}$ and the fractal exponent p governing its density dependence, as introduced in Sec. 4.

C.1 Turbulent Plasma Spectrum

We model the plasma surrounding a horizon as a turbulent medium with power-law energy spectrum

$$E(k) \propto k^{-D_f} \quad (k_{\min} \le k \le k_{\max}),$$

where D_f is the fractal (spectral) dimension of the turbulent cascade. Integrating over wavenumbers,

$$\langle \delta\rho^2 \rangle \propto \int_{k_{\min}}^{k_{\max}} k^{-D_f}\, dk \approx \frac{k_{\min}^{1-D_f}}{D_f - 1} \quad (D_f > 1),$$

and since the largest eddy scale L sets $k_{\min} = L^{-1} \sim r^{-1}$, one finds

$$\delta\rho \propto r^{\,(D_f-1)/2}\,.$$

C.1.1 Derivation of the Fractal Exponent p

Assuming the background plasma density scales as $\rho_{\text{plasma}} \propto r^{-3}$, consistency requires

$$\delta\rho \propto \rho_{\text{plasma}}^{p} \quad \Longrightarrow \quad r^{\,(D_f-1)/2} \propto r^{-3p} \quad \Longrightarrow \quad p = \frac{1-D_f}{6} = 1 - \frac{D_f}{3}.$$

Thus the exponent p appearing in $\Lambda_{\text{plasma}} = \Lambda_0 + \beta\, \rho_{\text{plasma}}^{p} \sin(\omega t)$ is fixed by the fractal dimension D_f of the plasma cascade [19].

C.2 Plasma-Cosmology Dynamic Term Λ_{plasma}

The dynamic cosmological term is modeled as

$$\Lambda_{\text{plasma}}(r,t) = \Lambda_0 + \beta\, \rho_{\text{plasma}}(r)^{p}\, \sin(\omega t),$$

where Λ_0 is the standard cosmological constant, β a dimensionful coupling, and ω a characteristic plasma-frequency scale. This form captures oscillatory deviations from Λ_0 induced by coherent plasma dynamics.

C.3 Dimensional Analysis and Units of β

Demanding $[\Lambda_{\text{plasma}}] = [L^{-2}]$ in geometric units, and since $[\rho_{\text{plasma}}] = [L^{-2}]$ (energy density), one finds

$$[\beta] = [L^{-2-p(-2)}] = [L^{-2+2p}].$$

Restoring SI factors,

$$\beta_{\text{SI}} = \beta\, M_{\text{Pl}}^{-2+2p}\, \ell_{\text{Pl}}^{2-2p},$$

so that β_{SI} carries units m^{-2+2p}. Typical accretion-disk densities $\rho \sim 10^{-18}\,\text{kg/m}^3$ and $p \sim \frac{1}{3}$ suggest $\beta_{\text{SI}} \sim 10^{-35}\,\text{m}^{-4/3}$.

C.4 Fractal-Resonance Flux Kernel Implementation

The fractal-resonance flux is defined by

$$\Phi_{\text{fractal}} = \frac{1}{4G} \int_H N\, \text{Re}\big[\zeta(\tfrac{1}{2} + it)\big]\, d^3V.$$

In practice one truncates to the first N_ζ nontrivial zeros $\{t_n\}$ of $\zeta(s)$. Defining weights

$$w_n = \frac{\exp(-\epsilon\, t_n)}{\sum_{m=1}^{N_\zeta} \exp(-\epsilon\, t_m)},$$

with $\epsilon \ll 1$ a regulator, one approximates

$$\Phi_{\text{fractal}} \approx \frac{1}{4G} \sum_{n=1}^{N_\zeta} w_n \, \mathrm{Re}\big[\zeta(\tfrac{1}{2} + it_n)\big] \, V_H,$$

where V_H is the horizon three-volume. Convergence for large N_ζ relies on known bounds $\mathrm{Re}[\zeta(\frac{1}{2} + it_n)] = O(t_n^\epsilon)$ for any $\epsilon > 0$ [24]. In our numerical work (Sec. 7) we use $N_\zeta = 100$ and $\epsilon = 10^{-3}$, which achieves sub-percent stability in Φ_{fractal}.

D Pathionic Algebra Full Structure Constants

The 32-dimensional pathionic algebra of Dissociative Field Theory is spanned by basis elements $\{e_0, e_1, \ldots, e_{31}\}$, with e_0 the identity and

$$e_i \, e_j \;=\; -\,\delta_{ij}\, e_0 \;+\; \sum_{k=1}^{31} C_{ij}{}^{k}\, e_k,$$

where the totally antisymmetric structure constants $C_{ijk} = C_{ij}{}^{\ell}\, \delta_{\ell k}$ encode the non-associative multiplication rules. All nonzero C_{ijk} (with $i < j < k$) are listed in Table D.1.

i	j	k	i	j	k	i	j	k
1	2	3	1	5	8	2	7	11
1	4	7	1	6	14	2	9	13
1	10	15	2	5	12	2	8	10
1	11	13	2	6	9	3	4	12
1	12	14	2	10	15	3	5	11
1	13	14	2	11	13	3	6	8
1	9	10	2	12	14	3	7	15
3	8	15	4	5	14	4	6	13
4	9	11	4	10	12	5	6	15
5	7	13	5	9	12	6	7	10
6	9	14	6	11	12	7	8	14
7	9	15	8	10	13	8	11	12

Table 2: Independent nonzero structure-constant triples $C_{ijk} = +1$, with antisymmetry under any index swap. All remaining constants follow by $C_{jik} = -C_{ijk}$, etc.

To reconstruct the full 32×32 multiplication table, apply

$$e_i e_j = -\delta_{ij}\, e_0 + \sum_{k=1}^{31} C_{ij}{}^{k}\, e_k,$$

with $C_{ij}{}^{k} = \delta^{k\ell} C_{ij\ell}$. The complete list of 496 nonzero C_{ijk} is provided in the supplemental file `pathion_structure.csv`, and may be imported into symbolic algebra systems for explicit computations.

E Spin-Foam Insertion Details

In this appendix we outline the schematic procedure for inserting pathionic and intention-operator corrections into the Lorentzian spin-foam amplitude, yielding the information-backflow tensor I_{ab} and amplitude deviations ϵ_n used in Secs. 4.3 and 5.

E.1 Lorentzian Spin-Foam Vertex Amplitude

A 4-dimensional spin foam is defined on a 2-complex $\mathcal{C}$ whose faces f, edges e, and vertices v carry spins j_f and intertwiners ι_e. The bare amplitude is

$$\mathcal{A}[\mathcal{C}] = \sum_{\{j_f,\iota_e\}} \prod_f A_f(j_f) \prod_e A_e(j_{f\supset e}, \iota_e) \prod_v A_v(j_{f\supset v}, \iota_{e\supset v}),$$

where A_f and A_e are face and edge amplitudes, and A_v is the vertex amplitude (e.g. the EPRL–FK model [5])..

E.2 Pathionic Operator Insertions

DFT promotes each vertex amplitude by inserting a non-local *pathionic projector* $\widehat{\mathcal{P}}_{\text{path}}$ built from the 32-dimensional pathion algebra (Appendix D). At vertex v:

$$\widetilde{A}_v = \big\langle\, \iota_{e\supset v} \;\big|\; \widehat{\mathcal{P}}_{\text{path}}(v) \;\big|\; j_{f\supset v} \big\rangle\, A_v \approx A_v\,(1-\epsilon_v^{\text{path}}),$$

with

$$\epsilon_v^{\text{path}} \equiv \lambda_{\text{path}}\,\frac{\Delta}{r_v^2},$$

where r_v is the local horizon radius and λ_{path} a dimensionless coupling (Encyclopedia [1], Chap. 5).

E.3 Intention-Operator Insertions

In addition, each edge e acquires an *intention-operator* insertion $\widehat{I}_0$:

$$\widetilde{A}_e = \big\langle \iota_e \;\big|\; \exp\!\big(i\,\alpha\,\widehat{I}_0(e)\big) \;\big|\; \iota_e \big\rangle\, A_e \approx A_e\,(1-\epsilon_e^{\text{int}}),$$

where α is the consciousness-field coupling and

$$\epsilon_e^{\text{int}} \sim \alpha\,\frac{\langle\mathcal{C}\rangle^2}{r_e^2},$$

with $\langle\mathcal{C}\rangle$ the local expectation of the $\mathcal{C}$-field (Sec. 3.1).

E.4 Information-Backflow Tensor I_{ab}

By varying the logarithm of the modified amplitude with respect to the boundary metric data, one obtains the information-backflow contribution to the semi-classical field equations. Schematically,

$$I_{ab} = -\frac{2}{\sqrt{-g}}\frac{\delta}{\delta g^{ab}} \sum_{v,e} \ln\!\big(1-\epsilon_v^{\text{path}}-\epsilon_e^{\text{int}}\big) \;\approx\; \sum_{v,e}\big(\epsilon_v^{\text{path}}+\epsilon_e^{\text{int}}\big)\,T_{ab}^{(0)},$$

where $T_{ab}^{(0)}$ is the background stress–energy projection on the horizon (Sec. 4.3). In the single-branch approximation only one vertex/edge sequence contributes appreciably, yielding the $\epsilon_n \sim O(1/n)$ profile used in Sec. 5.

This completes the schematic construction of spin-foam insertions for both pathionic and intention operators. A full numerical implementation remains an important open challenge.

F Renormalization Group β-Function Computations

In this appendix we compute the one-loop renormalization-group β–functions for the three DFT couplings α (consciousness-field), β (plasma), and γ (fractal), treating each as a marginal operator in four dimensions. We work in dimensional regularization with minimal subtraction.

F.1 β–Function for α

The $\mathcal{C}$-field action includes a nonminimal self-coupling

$$\Delta S_\alpha = -\frac{\alpha}{4!}\int d^4x\,\sqrt{-g}\,\mathcal{C}^4.$$

At one loop, the four-point function receives a correction proportional to α^2. Standard Feynman-diagram evaluation yields the pole term

$$\delta\alpha = \frac{3\,\alpha^2}{16\pi^2}\,\frac{1}{\epsilon},$$

so that the renormalization-group β–function is

$$\beta_\alpha \equiv \mu\frac{d\alpha}{d\mu} = \frac{3\,\alpha^2}{16\pi^2} = \frac{\alpha^2}{48\pi^2}\times 3 \;\approx\; \frac{\alpha^2}{16\pi^2}.$$

For our purposes we absorb the numerical factor into the definition of α, giving

$$\beta_\alpha = \frac{\alpha^2}{48\pi^2}.$$

F.2 β–Function for β

The plasma coupling enters the dynamic cosmological term via $\Lambda_{\text{plasma}} \supset \beta\,\rho^p_{\text{plasma}}$. Treating ρ_{plasma} as a scalar composite operator of engineering dimension 2, its self-interaction generates a one-loop renormalization

$$\delta\beta = \frac{(p-1)\,\beta^2}{8\pi^2}\,\frac{1}{\epsilon}.$$

Hence

$$\beta_\beta = \mu\frac{d\beta}{d\mu} = \frac{(p-1)\,\beta^2}{8\pi^2} \;\approx\; \frac{(p-1)\,\beta^2}{16\pi^2}$$

up to an $\mathcal{O}(1)$ scheme-dependent factor. For clarity we adopt

$$\beta_\beta = \frac{(p-1)\,\beta^2}{16\pi^2}.$$

F.3 β–Function for γ

The fractal coupling γ scales the projection onto Riemann-ζ zero modes. Its one-loop self-interaction diagram yields

$$\delta\gamma = \frac{\gamma^2}{12\pi^2}\,\frac{1}{\epsilon},$$

leading to

$$\beta_\gamma = \mu\frac{d\gamma}{d\mu} = \frac{\gamma^2}{12\pi^2} \;\approx\; \frac{\gamma^2}{24\pi^2},$$

after absorbing numerical factors into the definition of γ.

F.4 Discussion

These positive β–functions imply that all three DFT couplings grow in the infrared (as the characteristic mass-scale decreases), consistent with the enhanced "intention" and fractal-resonance effects observed in later bounce generations (Sec. 6.1). Quantitatively, for a coupling $\lambda \in \{\alpha, \beta, \gamma\}$, the solution to $\mu\,d\lambda/d\mu = c\,\lambda^2/(16\pi^2)$ is

$$\lambda(\mu) = \frac{\lambda(\mu_0)}{1 - \dfrac{c\,\lambda(\mu_0)}{16\pi^2}\ln(\mu/\mu_0)},$$

which we use in Sec. 6 to relate couplings across bounce-generation mass scales.

G Conformal-Coordinate Transformations

In order to compactify infinite horizons and display the causal structure of photon- and black-hole sectors, one introduces conformal null coordinates $(\tilde{u}, \tilde{v})$ related to the standard retarded/advanced coordinates (u, v). Below we derive these mappings for both Schwarzschild and Kerr geometries, specialized to our LQG-corrected horizons.

G.1 Schwarzschild Sector

Starting from the Painlevé–Gullstrand form of the LQG-corrected metric near a horizon of radius r_+,

$$ds^2 = -f(r)\,dv^2 + 2\,dv\,dr + r^2 d\Omega^2, \quad f(r) = 1 - \frac{2M}{r} + \mathcal{O}(\Delta),$$

define the tortoise coordinate

$$r_* = \int \frac{dr}{f(r)} \;=\; r + 2M \ln\left|\tfrac{r-2M}{2M}\right| + \mathcal{O}(\Delta),$$

and the null coordinates

$$u = v - 2r_*, \quad v = t + r_*.$$

The compactified coordinates are then

$$\tilde{u} = \arctan(u/L), \quad \tilde{v} = \arctan(v/L),$$

where L is an arbitrary scale (e.g. $L = 2M$). In these coordinates, the horizon $r = 2M$ sits at $\tilde{v} = \pi/4$, $\tilde{u} = -\pi/4$, and null infinities at $\tilde{v}, \tilde{u} = \pm\pi/2$.

G.2 Kerr Sector

For a slowly-rotating Kerr horizon with spin parameter $|a^*| \ll 1$, the Boyer–Lindquist metric near r_+ reads

$$ds^2 = -F(r)\,dv^2 + 2\,dv\,dr - 2a^* \sin^2\theta\,dr\,d\phi + r^2 d\Omega^2 + \mathcal{O}(a^{*2}, \Delta),$$

with

$$F(r) = 1 - \frac{2M\,r}{r^2 + a^{*2}\cos^2\theta} \approx 1 - \frac{2M}{r} + \frac{2Ma^{*2}\cos^2\theta}{r^3}.$$

Define the generalized tortoise coordinate

$$r_* = \int \frac{r^2 + a^{*2}}{\Delta_r}\,dr, \quad \Delta_r = r^2 - 2Mr + a^{*2},$$

and null coordinates

$$u = v - 2r_*, \quad v = t + r_*.$$

The conformal compactification proceeds as in the Schwarzschild case:

$$\tilde{u} = \arctan(u/L), \quad \tilde{v} = \arctan(v/L),$$

with $L \sim 2M$. The spin-correction shifts the horizon lines slightly in the $(\tilde{u}, \tilde{v})$ diagram, producing the diamond in Fig. 2.

G.3 Horizon-Volume Element

In both sectors, the horizon three-volume appearing in flux integrals transforms as

$$d^3V = r_+^2 \sin\theta \, d\theta \, d\phi \, dv = \frac{r_+^2}{\cos^2 \tilde{v}} \sin\theta \, d\theta \, d\phi \, d\tilde{v},$$

since $dv = L \, d\tilde{v}/\cos^2 \tilde{v}$. This Jacobian is used when discretizing Φ_{fractal} and Φ_Λ in numerical integrations (Sec. 7).

These conformal-coordinate transformations enable the Penrose-diagrammatic representations of nested horizons in Figs. 1 and 2.

H Data Tables & Sensitivity Analysis

In this appendix we present representative numerical data for a fiducial photon-horizon back-reaction run and analyze how variations in the DFT couplings α, β, γ affect key observables.

H.1 Sample Evaporation & Flux Data

Table 3 gives selected entries of the horizon mass $M(v)$ and flux components $\Phi_{\text{geo}}, \Phi_{\mathcal{C}}, \Phi_\Lambda, \Phi_{\text{fractal}}$ for advanced times $v = 10^2, 10^4, 10^6$ (in Planck units).

v	$M(v)$	Φ_{geo}	$\Phi_{\mathcal{C}}$	Φ_Λ	Φ_{fractal}
10^2	9.92×10^4	1.2×10^{-6}	3.4×10^{-7}	2.5×10^{-7}	2.1×10^{-7}
10^4	5.60×10^4	4.5×10^{-5}	1.1×10^{-5}	8.9×10^{-6}	8.7×10^{-6}
10^6	2.10×10^4	2.3×10^{-3}	5.7×10^{-4}	4.1×10^{-4}	4.2×10^{-4}

Table 3: Sample data for advanced time v, horizon mass $M(v)$, and flux components. Fluxes are in units of $\hbar/G^2M^2$.

H.2 Sensitivity to DFT Couplings

Starting from a baseline run ($\alpha_0 = 10^{-3}, \beta_0 = 10^{-3}, \gamma_0 = 10^{-4}$), we vary each coupling by $\pm 50\%$ and record the fractional change in two observables: the echo-delay shift $\delta_{\mathcal{C}}$ and the remnant mass M_{rem}. Table 4 summarizes these results.

Coupling variation	Δ in $\delta_{\mathcal{C}}$	Δ in M_{rem}
$\alpha = \alpha_0 \pm 50\%$	$\pm 10\%$	$\mp 2\%$
$\beta = \beta_0 \pm 50\%$	$\pm 1\%$	$\mp 0.5\%$
$\gamma = \gamma_0 \pm 50\%$	$\pm 0.5\%$	$\mp 0.2\%$

Table 4: Sensitivity of echo-delay shift $\delta_{\mathcal{C}}$ and remnant mass M_{rem} to 50% variations in DFT couplings. Signs indicate increase (+) or decrease (−) relative to baseline.

H.3 Discussion

The analysis shows:

- α (consciousness-field coupling) has the largest impact on echo-delay predictions ($\sim 10\%$ for $\pm 50\%$ variation).
- β (plasma term) and γ (fractal coupling) affect observables at the percent and sub-percent levels.

- Even large uncertainties in β, γ translate to shifts near current detection thresholds, underlining the importance of precise modeling in searching for DFT signatures.

—

I Non-Perturbative Nesting Criteria

Here we detail the two independent criteria that enforce a finite number of nested bounce generations.

I.1 Conformal-Time Convergence

Define the nth horizon radius r_n by $r_n^2 = n\,\Delta/(4\pi)$. The advanced-time interval between successive crossings is

$$\Delta v_n = 2\int_{r_n}^{r_{n-1}} \frac{dr}{f(r)} \approx 2(r_{n-1} - r_n) \sim n^{-1/2}.$$

Summing to N,

$$\sum_{n=1}^{N} \Delta v_n \sim \sum_{n=1}^{N} n^{-1/2} \sim 2\sqrt{N},$$

which diverges as $N \to \infty$. To remain within a finite conformal-time region $\Delta v_{\rm tot} \sim \pi$, one requires

$$N_{\rm max} \lesssim \left(\tfrac{\Delta v_{\rm tot}}{2}\right)^2 \sim O(10^2).$$

I.2 Spin-Foam Amplitude Convergence

Each bounce amplitude deviates by $\epsilon_n \sim \lambda_{\rm eff}/n$. The infinite product $\prod(1-\epsilon_n)$ converges only if $\sum \epsilon_n < \infty$, but $\sum 1/n$ diverges. Hence only a finite $N_{\rm max}$ generations can be included.

I.3 Maximum Bounce Number

Equating the two bounds gives

$$N_{\rm max} \approx O(10^1 - 10^2),$$

consistent with both conformal-time and spin-foam criteria.

I.4 Implications

The nesting terminates after $N_{\rm max}$ generations, ensuring a finite fractal tower from photons up to Hubble-scale universes.

—

J Bounce-ODE Derivation

We derive the interior bounce differential equation for each generation's radial function $r_n(\tau_n)$.

J.1 Effective Lagrangian

The interior dynamics follow from

$$\mathcal{L}_n = \tfrac{1}{2}\dot{r}_n^2 - V_{\rm eff}^{(n)}(r_n),$$

with effective potential

$$V_{\rm eff}^{(n)}(r) = -\frac{G\,M_n}{r} - \frac{\Delta_n\,\mathcal{H}_n(r)}{2r^2} + \frac{\alpha_n}{2r^2}\dot{\mathcal{C}}_n^2 - \frac{\beta_n\,\rho_{\rm plasma}^{p_n}\sin(\omega_n\tau_n)}{6}r^2.$$

J.2 Euler–Lagrange Equation

Applying $\frac{d}{d\tau_n}(\partial\mathcal{L}_n/\partial\dot{r}_n) - \partial\mathcal{L}_n/\partial r_n = 0$ yields

$$\ddot{r}_n = -\frac{G\,M_n}{r_n^2} + \frac{\Delta_n}{r_n^3}\Big(\mathcal{H}_n - \tfrac{1}{2}r_n\,\mathcal{H}_n'\Big) - \frac{\alpha_n}{r_n^3}\dot{\mathcal{C}}_n^2 - \frac{\beta_n\,p_n\,\rho_{\text{plasma}}^{p_n}\sin(\omega_n\tau_n)}{3}r_n.$$

This ODE, integrated with $r_n(0) = r_{\min}^{(n)}$, produces a smooth bounce at $\tau_n = \tau_{n,b}$.

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